Project documentation and wiki development for iGEM projects

Contents

Published by Liliana Sanfilippo
liliana.sanfilippo@uni-bielefeld.de

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Introduction

This guide is a comprehensive introduction to creating and managing an iGEM wiki from the absolute basics to the technical documentation of a modern React-based template wiki designed specifically for iGEM teams and to both seasonal project documentation for the competition as well as how to document for future teams from your university.

It is organized into various sections that cover specific aspects of project management, collaboration, design, documentation, and technical implementation.

Throughout this guide, I will assume a successful activation of the team Wiki but will walk the reader through the fundamental aspects of setting up the Wiki, as well as providing a general overview of project documentation. I will emphasize key elements that are essential for the Wiki team, such as effective documentation, clarity, and organizational strategies. The coding section will specifically focus on React applications, explaining the basics and offering guidance on best practices and methods for creating a dynamic and user-friendly interface.

At the end of the guide, there will be an index list to help you easily find specific topics and navigate through the content more efficiently.

This guide also serves as documentation to the associated template wiki, which teams are free to use as a starting point. It can be found at `https://github.com/liliana-sanfilippo/igem-wiki-guide`.

The Template Wiki is a React-based, easily customizable template for iGEM wikis. It is aimed at first-time teams with little experience as well as advanced iGEM teams. The goal is to make it easier to get started with the technical implementation of your wiki without having to compromise on functionality or flexibility.

The template is currently developed further, including some independent packages. It offers a modern, modular code base that can be easily adapted and extended. The aim is better usability and maintainability compared to the example wiki provided by iGEM through additional functions, including components designed to work with the template specifically.

It is a first version and I will be working on improving it, so any feedback or ideas can be sent to liliana.sanfilippo@uni-bielefeld.de or raised as issues on GitHub.

Documentation for other tools or packages by myself is not included, but wil be linked.

While updating the guide, some already finished sections will be re-added and the following aspects will be added:

- An in-depth explanation on how to implement a citation React component
- Setting Global Variables in React
- Wiki Thaw
- Utilizing Bootstrap
- FAQ
- Merge Conflicts
- Utilizing ChatGPT
- Where can I get help?
- Definitions of terms such as CSS path, CSS selector
- Errors and Troubleshooting
- html link targets
- Further Collaborations and Considerations with other subteams (e.g. Sponsoring and logos)
- Scrolling animations
- Possibly education and HP documentation frameworks
- Appendix for first time teams with basic, but possibly unknown information and considerations

As well as additional images outlining the instructions and examples given.

Documentati

Chapter 1

Setting up the documentation and organisation

Creating well-structured documentation is essential for effective communication and information retention. It is also very helpful for the next team at your university to have good internal documentation of contacts and resources.

We will start with the internal documentation and continue with the project documentation for the competition. Please adapt the suggestions to your needs.

1.1 Internal and general documentation

A lot of software solutions and methods for documentation exist. I will not discuss the advantages and disadvantages of these, but instead mention general considerations. The first and foremost being: Your documentation system has to be thorough, but easy to use.

The most sophisticated system is of no use, if half the team cannot use it on their device or it takes weeks to learn it.

1.1.1 Meeting protocols

In Bielefeld, we have a meeting template - that varies each year - for team meetings. Team members are asked to prefill it. This also opens up the option to put town tops anys time, lessening the amount of forgotten points. It usually contains:

- A “Top” section. Tops being discussion points where input or decisions are needed.
- An “Info” section that is not read out. Everyone is responsible for reading it themselves as it contains information relevant to the whole team.
- A “Subteam update” section that can later be used to lookup information.

Further information such as who was present or who wrote the protocol can be added.

Subteam meetings are usually not as formally documented, but protocols are expected.

The minute taker changes every meeting as we rotate through a list to render discussions about who has to do it unnecessary.

It is advisable to have meeting protocols from the beginning. Depending on your team structure, experience and regarding time management, you should consider:

- **Preparing agendas**: Starting meetings with a pre-prepared agenda outlining the topics for discussion.
- This helps keep the meeting focused and provides a structure for your notes.
- **Tracking decisions**: Document any decisions or agreements reached during the meeting to provide a clear record of team choices.
- **File Storage and Accessibility:** The meeting notes should be accessible to the whole team afterward and multiple versions should be most urgently avoided.
- **Summaries**: Some teams may find it helpful to create meeting summaries after the fact. Though this can also be a time drain.

1.1.2 Task tracking

While there might not be as many tasks in the beginning I urge you to use some kind of tracking system. Many apps are only available for certain browsers or OS, therefore we usually use an excel sheet. I recommend writing down not only the task and the assignee, but also a deadline.

That way you can check in with each other, if tasks get lost. Please remember to be kind with your team mates and that this is not about control or who is right, but about being thorough.

Do this also for texts or other wiki content! In regard to texts and wiki content, we have an excel sheet containing:

- Topic / Title
- Author(s)
- Wiki page it will be published on
- Category / Subteam

- Deadline
- Done?
- Three columns where other team members put their initials of they proofread the text
- A column where an advisor puts their initial if they proofread the text
- A checkbox for the author to check if they have reworked the text according to the proofreading and corrections / suggestions
- Is the text on the wiki?

This allows for automatic counting of how many texts / proofreading per person were done to avoid having like two people do all the texts alone.

1.1.3 Sponsoring

If you get sponsoring from companies, it is likely you promise them things in return or even make contracts. For example presenting their logo on your wiki.

To keep contacts over the years, you need to not document the companies, but the contact data, who you spoke with and possible additional info such as rules or recommendations they gave you.

To avoid looking unorganized and unprofessional, I recommend writing down who contacted the companies (if they have been contacted at all) and the status of the outreach.

In Bielefeld, we use an excel sheet for all sponsoring contacts and a second sheet to document the won sponsors.

The general sheet contains:

- Company name
- Person in charge
- Status
  - First outreach (no answer yet)
  - Second outreach (no answer yet)
  - Not started
  - Declined
  - In contact
  - Won

- Contact person (if there is one)
- Phone number (if available)
- Email
- Additional info / notes (e.g. “”, “prefers high social media activity”)
- Products (they could possibly provide)

The sheet for won sponsors includes (besides company name, etc.):

- Money sponsoring (if given)
- Commodity sponsoring (if given)
- Sponsoring status
  - To be shipped
  - Not transferred yet
  - Received
- Contract?
- Category (if you use this system)
  - None
  - Bronze
  - Silver
  - Gold
- Did they provide the logo?
- A column for each benefit you can provide the companies where you note if they shall receive it and if you fulfilled that commitment. This can for example include:
  - Logo on wiki
  - Logo on team shirts
  - Social media appreciation post

1.1.4 Images and videos

Applications such as Microsoft Teams struggle with a high load of videos or images. Consider using a cloud or shared university file server.

I recommend to document if you have the consent of the people in the images and videos if you plan on releasing footage of external people.

1.1.5 Attributions

Do not wait until the end to document your team attributions or external attributions. Do so on a rolling basis and make sure to explicitly distribute the responsibilities. Otherwise, the subteam might end up thinking the organization team does the attributions and the other way around, and they do not get done.

I recommend documenting them the way you have to submit them later on directly.

1.1.6 HP contacts

Similar to the sponsoring contact, document:

- Name
- Institution / Affiliation
- Contact info
- Person in charge of contact
- Outreach status (“First outreach”, “Declines”, etc.)

Additionally, document:

- Stakeholder category
- Did they consent to you using their picture
- Did they consent to publication of interviews / transcripts or do they want to read ot first, ...

Consent Management includes explaining how the “freezing” of iGEM Wikis works and that it will not be possible to change or remove information after the project ends. Be mindful to receive feedback for quotes and transcripts of interviews. Especially if you need to translate conversations to English, the stakeholders should get the chance to comment and, if necessary, correct your translations.

I am diving further into HP documentation in the section Documentation for the competition.

1.1.7 Staying on top of the documentation

See also: Internal freeze approach

1.1.8 Milestone approach

In 2025, I tried the approach to set specific dates for check-ins with the team at the beginning of the wiki work, akin to agile project management (but less micromanager-y, no stand-up meetings or daily scrums or whatever). The team had specific design and documentation task that were to be finished and the expectation was that what was not finished at the deadline had to be caught up on during co-working sessions. After the design phase, biweekly checkins (and working sessions) were established with specifications in what kind of product the various tasks should have.

We are now continuing this practice in 2026.

Tasks can be specific or abstract such as "the first draft for every wiki text for HP work up until this point is done" or "every text from the last intervall is proofread".

Some abstract tasks we implemented are:

- All existing (and proofread) texts (and graphics where applicable) from the last iteration are on the wiki (*HTML / Template*)
- Texts including sources exist for all past events and interviews (*Word (texts) / BibTeX (sources)*)
- All existing graphics and extra data have been uploaded to the upload tool (*upload*)
- Engineering cycle up to this point (outline) with Lab Team and including HP (*Word or PowerPoint*)

This works quite well and the team was usually up to date with their documentation.

1.2 Documentation for the competition

1.2.1 General considerations

Here are some general considerations for documentation, including various methods to create PDFs and layout decisions for protocols and notes.

Purpose and Audience

Clearly define the purpose of the document(ation) and be aware your target audience. Understanding the audience's knowledge level will guide your writing style and content depth. Of course judges will most likely come from a biotech background, but they can still be unfamiliar with niche topics. A different

example where your documentation and documents need audience considerations would be flyers or videos for educational purposes, especially if you want to choose education as a special prize.

Clarity and Consistency

Use clear, concise language to convey information effectively. Maintain consistency in terminology, formatting, and style throughout the document to enhance readability and comprehension. There is no special prize for using the most complicated words or coming up with the most intricate way to describe simple procedures.

Sustainability

Decide on styles and approaches early on and think about using English from the get-go to avoid having to translate and rework your notes and protocols later on. Remember that you need to credit translators and potentially translating tools in the attributions.

Structure and Organization

Use a logical structure with a clear hierarchy, even if it seems basic or unnecessary to you.

- **Title Page:** Include the document title, author(s), date, and possibly your logo. Remember, the document is presented on your wiki, but it is possible it will be distributed in a different context, and therefore you should always add all necessary information to each document.
- **Table of Contents:** Do not forget to provide a navigation aid for longer documents. If possible, create a table of content with hyperlinks.
- **Structure** Divide content into clear sections with appropriate headings. Possibly use a glossary for intricate documents.
- **Conclusion:** Summarize key points to ensure the reader understands the main points and offer recommendations if applicable.

Choosing a Format for Documentation

Word Processing Software (e.g., Microsoft Word, Google Docs): These tools allow for straightforward document creation with various formatting options. You can export your documents as PDFs easily, ensuring compatibility across different devices and allowing collaborative writing.

LaTeX: Ideal for more complex documents, LaTeX offers precise control over formatting, making it perfect for scientific papers and technical documentation. It is particularly beneficial for documents with mathematical equations, citations, and references. Services such as Overleaf allow collaborative writing.

It is possible to create PDF-masks with design tools such as Canva, if you want to add further styles or branding to your documents.

Even if you are unsure what to use yet, a digital text file of any sort is always easier to edit and transfer than handwritten notes!

1.3 Specific considerations

I recommend establishing a documentation framework for every subteam/topic. This ensures you start every endeavour with an idea what you want out of it, the ability to compare outcomes and actually already a way to tidily present everything on the wiki!

See Documentation framework examples for some examples.

It might seem like you are just going through the motions instead of truly diving into the topics. But in my opinion that is not true. You use these frameworks as a starting point and creating them forces you to think about what your goals are, what you expect and what is important to document.

Picture this: You are filling out the questionnaire/form for an expert human practice encounter. You filled out the basic information and reflected on the outcomes and how to integrate this into your project. There is another piece of information you cannot really fit into the form, because it does not ask for it. That is an insight. What did you not consider that lead to this oversight? What can you retrospectively say about this aspect in regard to the other encounters?

And always remember: No answer is also information. If for example an expert says that your question about safety does not have a (good answer), because that safety issue is not applicable, write. that. down.

Further, using structured documentation like this allows you to use JSON-like data for reusable wiki components, allowing a certain degree if autonation.

1.3.1 Lab

Well-maintained lab books are essential for tracking progress, reproducing experiments, and ensuring accountability. The following aspects seem basic and natural which unfortunately puts them at risk of being forgotten for this exact reason.

- **Digitalize**: Most projects get busier towards the end, and it is unlikely you will have the time to digitalize your notes later on. It is best to keep it digitally from the beginning, in whatever exact form. Even a `.txt` file is better than handwritten notes, because the text can simply be copied to the wiki in the worst case.
- **Backup**: Regularly back up entries to prevent data loss.
- **Raw Data**: Always include raw data alongside processed or analyzed data and save the data separately, too.

If you want to use a lab book software, please check beforehand if it allows you to export your (raw) data and notes in a useful way. Usually, useful ways are easily processable data files such as `.cvs` or `.json`.

Protocol design examples

- JU Krakow 2024

1.3.2 Integrated Human Practices

Documenting Integrated Human Practices involves tracking contacts, conversations, permissions, and the use of any media, ensuring compliance with ethical and legal standards.
Both to avoid unnecessary work and to maintain a professional demeanor towards your stakeholders. See also HP contacts regarding consent management.

- **Tracking Contacts and Interactions**:
  - Maintain a list of **Contact Information** of the individuals and institutions you both want to contact and already contacted. Include details such as *role, affiliation, date of first contact,* etc.
  - Keep track of **Who Contacted Whom** to avoid contacting the same person multiple times. This helps in maintaining accountability and tracking follow-ups.
- **Recording Conversations and Outcomes**
  - If possible, clear the **Type of Documentation** with the interviewee beforehand.
  - Try to organize **High Quality Tools** to record and test them beforehand.
  - Be aware of background noises and keep in mind that the videos could be useful for your project presentation video.
  - Create **Meeting Summaries** including *main topics*, *key takeaways*, *quotes* and *to-dos*.

- **Categorizing Stakeholders and Their Input**
  - The **Categorization of Contacts** into categories such as *Academia, Industry* or *Community* should be started as soon as possible to streamline documentation and to facilitate references back to relevant discussions.
  - Keep track of the **Implementation of Advice and Input** you received to be able to cross-reference from you Integrated Human Practice to other aspects of your project.
- **Ethical Considerations**
  - Ensure that no **Sensitive Information** is included in your published material. This can include *confidential project data* or *compromising information.* If necessary, censor details to ensure the **Privacy** of individuals such as patients, children or other vulnerable stakeholders.
  - Maintain proper **Attribution** for both intellectual input and media usage.
- **Maintaining Transparency**
  - Maintain **Transparency in Reporting** by including clear and concise references to your interactions with stakeholders when writing Wiki texts.
  - Be upfront about **Changes and Censorship** in your documentation.

1.3.3 MeetUps

When documenting an iGEM meetup, it's important to capture not only the logistics but also the valuable exchanges, collaborations, and outcomes from the event.

- **Photos and Videos**: Be sure to organize media documentation beforehand and take pictures of key moments. Not only for your team but to support the attending teams and allow them to concentrate on the event.
- **Expert Feedback**: If experts or judges were present, document the feedback they provided to your team or others. Highlight how this feedback can influence your project moving forward.
- **Material Exchange**: If any teams shared protocols, tools, or resources during the meetup, keep track of what was exchanged and which team shared it.
- **Lessons Learned**: Capture insights or takeaways from the event that could influence your project, approach, or team dynamics moving forward.
- **Online Resources**: If presentations, slides, or meeting recordings were shared, document how these materials can be accessed later (e.g., through a shared drive or link).
- **Meetup Agenda**: Include the event's agenda or a summary of the planned sessions, presentations, or workshops. Be sure to note changes.
- **Feedback**: If possible, gather feedback on-site at the end of the event to ensure high involvement and profit from fresh impressions.

Wiki preparation and planning

Chapter 2

Design

2.1 Collaboration with the Creativity/Design Team

To create an effective and visually appealing iGEM Wiki, collaboration with the Creativity/Design team is essential. You do not only want to simply collect information on your wiki, but present it in a coherent, readable and convincing way to both fellow iGEMers and judges. Consider the following design aspects:

Choosing Colors: Select a cohesive color palette that reflects your team's branding.
If the team has no prior experience with design, color psychology[**?**] or simply color palette choosing tools can be a helpful starting point, especially in small teams with limited resources.

Initial Design Work: Ideally, start in a design program or layout tool (e.g. Figma) to visualize your ideas before implementing them on the Wiki. Alternatively, vision boards or analogue planning methods such as drawing work as well.

Choosing Illustration Styles: Decide on illustration styles that align with your overall design and branding. Consistency in style helps create a unified look across your Wiki, making it more visually appealing and easier for users to navigate.
Style considerations can be the software used for illustrations and specific details such as the usage of borders and shadows for in-house designs.

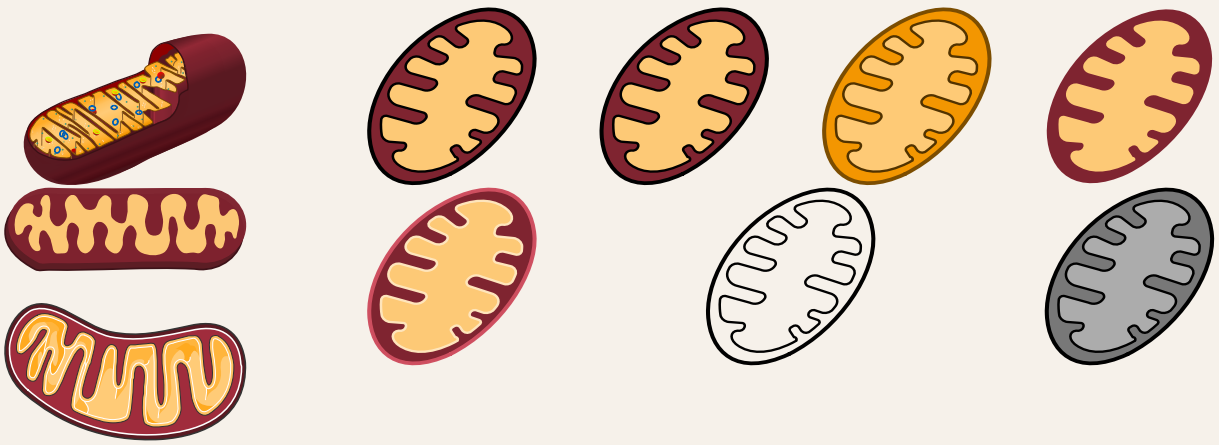


Figure 2.1: Different styles.

Logo usage: There are many possibilities to use your team logo on your wiki. Both for simple things such as buttons as well as more complex things such as animations, progress bars or a guide through your website.

2.2 Design Parameters

- **Typography:** Prioritize typography to enhance readability and aesthetics.
  - Maintain appropriate line length and spacing for readability.
  - Use standardized font sizes for consistency across the Wiki.
- **Simplicity:** Aim to simplify designs for a cleaner look.
  - Limit to 2–3 headings and colors to maintain clarity and focus.
  - Minimize the number of design parameters to streamline the design process.
  - Use colors purposefully; avoid unnecessary embellishments unless justified.
- **Consistency:** Unify illustration styles and design elements for a cohesive appearance.
  - Ensure consistent image formats and alignment (preferably left-aligned, avoiding justified text).
  - Align infographic styles to maintain a cohesive visual narrative.
- **Bootstrap Values:** Consider overriding Bootstrap defaults as needed for your design (See Bootstrap).
- **Call to action and clear navigation:** Include a welcoming call to action on the homepage, using colors meaningfully to guide users.

  - Possibly add further guidance such as highlights, to guide the user through the wiki and help the judges find all necessary information.
  - Use a clear menu structure and possibly breadcrumbs.
  - Avoid deep nesting your pages and aim for a flat, user-friendly structure. You want the judges to easily find all you pages from the standard URLs.

- **Content Organization:** Clear information hierarchy helps users find what they're looking for faster.
  - Utilize cards or expandable sections for content organization.
  - Ensure the basic information is always available and additional information is, while easily locatable, tucked away and not distracting the reader.

- **Responsiveness:** Ensure that all design elements adapt smoothly to different screen sizes (See Bootstrap).
  - Prioritize mobile usability since people may want to look at your wiki on their phones during or after presentations and poster sessions.
  - Verify that navigation, text, and visuals remain accessible and visually consistent across devices.

- **Accessibility:** Design for inclusivity by ensuring content is accessible to all users. Especially if you aim for the inclusivity special prize.

- See: Accessibility - Under Construction
  - Maintain sufficient color contrast for readability.
  - Use semantic HTML and alt text for images.

2.3 Examples and resources from teams

Color design guide by MIT-Mahe 2025: `https://static.igem.wiki/teams/5645/wiki-pdfs/clarity-by-design.pdf`.

Illustration style examples:

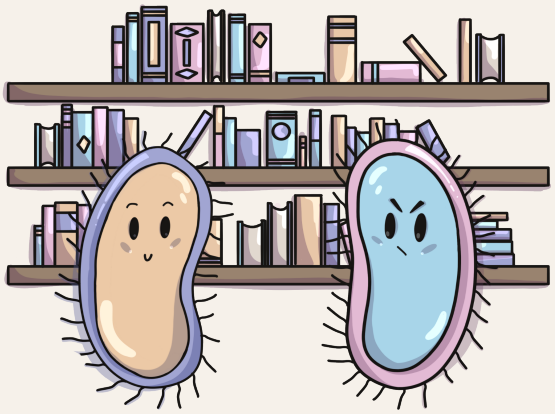


Figure 2.2: Illustration by Insa ENS Lyon 1 in 2023: `https://2023.igem.wiki/insaenslyon1/human-practices`.

Figure 2.3: Illustration by EPFL in 2024: `https://2024.igem.wiki/epfl/`.

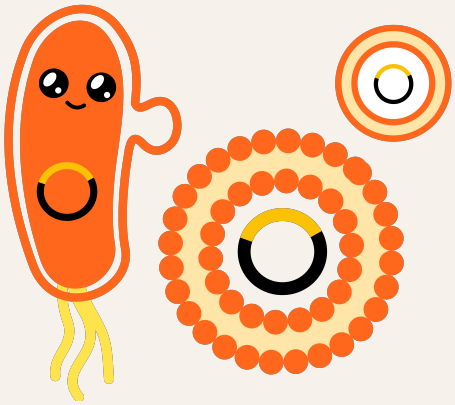


Figure 2.4: Illustration by Freiburg 2024: `https://2024.igem.wiki/freiburg/description`.

Chapter 3

User-friendliness and usability - Under construction

In the meantime a few examples:

3.1 Links

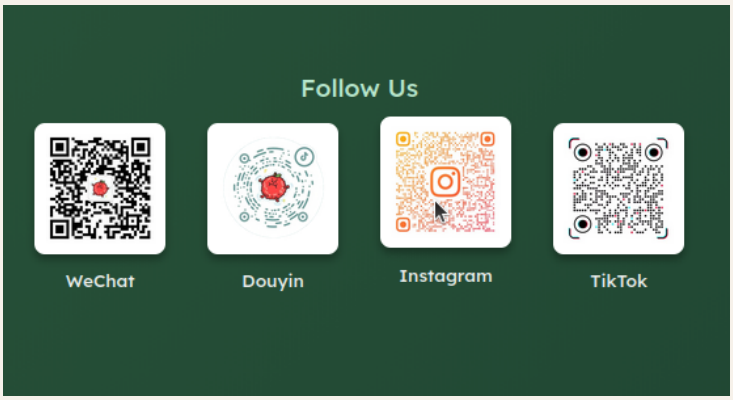


Figure 3.1: Having QR codes to lin socials might seem like a good idea, but you should always link them as well. In this example, the code images do not even open full screen or bigger on click so a user can scan a soecific one. From `https://2025.igem.wiki/hkust-gz/`

Chapter 4

Website and page structures

4.1 Standard URL Pages

iGEM provides standard URL pages that partly correspond to deliverables required for medal eligibility. These pages should be carefully curated to meet specific criteria, and judges will often look at them first. Here are the essential pages you must include:

Bronze Medal Criteria

- **Attributions:** This page outlines the contributions of each team member and any external help you received.
  - *https://year.igem.wiki/example/attributions*
- **Project Description:** A detailed description of your project, summarizing the scientific question you are addressing.
  - *https://year.igem.wiki/example/description*
- **Contribution:** Document how your team contributed to the iGEM community, whether through improved protocols, new parts, or educational resources.
  - *https://year.igem.wiki/example/contribution*

Silver Medal Criteria

- **Engineering Success:** Outline how your project follows the engineering design cycle, including design, build, test, and learn phases.
  - *https://year.igem.wiki/example/engineering*
- **Human Practices:** Demonstrate how your project interacts with and is shaped by the broader societal, ethical, and environmental context.
  - *https://year.igem.wiki/example/human-practices*

Gold Medal Criteria

For Gold, your team needs to document work for Special Prizes. The structure of these pages may vary depending on the prizes you select.
These standard URL pages are automatically linked to your Judging Form, so it's essential that all relevant achievements are documented here to ensure proper judging and recognition.

4.2 Additional Pages and Considerations

While the Standard URL Pages cover core requirements, your team may need to include additional pages to give a full picture of your project. Consider adding separate pages for the following:

- **Team Page:** Introduce the team members, their roles, and their contributions. This humanizes your project and gives credit to everyone involved.
- **Lab Books / Notebook:** Document your experiments and lab work. Each experiment or set of related experiments can have its own section for better clarity.
- **Meeting Documentation:** Keep track of all important team meetings, especially those related to project decision-making, collaborations, and interactions with stakeholders.
- **Sponsors and Partners:** List all your project sponsors, partners, and external supporters. This can be a good way for acknowledging specific contributions and maintaining transparency. If you have official sponsoring contracts, you can offer the sponsor page as a way of advertising to your sponsors.
- **Materials and Methods / Experiments:** Naturally, you need to provide detailed information about the techniques, equipment, and protocols used in your project for both judges and future teams.
- **Results:** Summarize your findings and data analysis to give the user overviews and to offer easy entry into every aspect of your project. Depending on your results' complexity, this may warrant its own section or multiple pages.
- **Parts:** Even if you are not going for a part prize, list the biological parts you designed or used in your project, following iGEM's standards for documenting parts.
- **Sustainability:** Include this page to highlight how your project contributes to sustainable development goals or impacts the environment positively.

Some teams like to spread out their documentation over non-standard pages. Feel free to do this but make sure to link everything on the standard pages!

Further pages I encountered are:

- Judging / Awards
- Outlook / Future Directions / Application
- Proof of Concept
- Outreach
- Implementation
- Plasmid Design
- Design
- Brand / Development of logo design etc.
- Collaboration
- Milestones / Highlights
- Gallery
- Special Thanks
- Pages for specific side projects such as a video tutorial series, Social Research / Survey, etc.
- Glossary

4.3 Structuring for Easy Navigation

It's important to ensure that every aspect of your project can be reached through the standard URL pages. For example, your *Human Practices* page can link to more detailed reports or outcomes in related sections such as *Sustainability* or *Ethics*, but make sure the essential content is summarized within the *Human Practices* page itself.
To ensure easy navigation and avoid clutter, consider using clear menus and submenus that guide users through different sections. Items such as *Proof of Concept* or *Design* can be standalone pages but should also be cross-referenced in key standard pages such as *Engineering Success* to ensure judges can find relevant information quickly.

4.4 Enhancing the Navigation Bar

The navigation bar is one of the most crucial elements of your wiki, guiding users through the content and making sure they can quickly find what they need. Here are some key considerations to ensure your navigation bar is both functional and user-friendly:

- **Linking Sections, Not Just Pages:** It's important to know that your navigation bar doesn't have to link only to full pages. You can also link directly to specific sections within a page. This is

especially useful if you have long pages with different subtopics. For example, in a dropdown menu for "Project", you could link to the "Design" section of the main project page without needing to create a separate page for it. This saves time and avoids unnecessary fragmentation of content while ensuring that all important sections are easily accessible.

```
{
name: "Project Description",
title: "Project Description",
path: "/description?scrollTo=Abstract"
}
```

- **Creating a 'Highlights' Dropdown:** A smart way to organize your navigation bar is by adding a Highlights dropdown, which can give users a quick tour of the key content. This is ideal for judges or visitors who want to get an overview without navigating through each section individually. Include the most critical pages like:
  - Project Description
  - Human Practices
  - Engineering Success
  - Contribution

  This offers a streamlined navigation experience, allowing users to quickly jump to the core of your project.
- **Page and Folder Highlighting:** For enhanced user experience, it's a good idea to highlight the current page or folder that the user is on within the navigation bar. This visual cue helps users understand where they are within the site structure. Many modern websites use a simple color change or underline to show which menu item is active. This is especially useful when users are navigating through multiple pages or sections, preventing them from getting lost in a large wiki structure.
- **Dropdown Menus for Section and Page Links:** Dropdowns can help organize both pages and sections under a single heading. For example, if you have a "Project" dropdown, it can contain links to separate pages like *Project Description*, while also linking directly to sections of a single page, like *Design* or *Results*. This ensures that teams don't feel pressured to create separate pages just to mention something on the navigation bar—allowing a better balance between organization and simplicity.

Ensure that your navigation bar includes direct links to all key sections such as *Project Description*, *Engineering Success*, and *Human Practices*.

Also consider:

- **Clear Labeling:** Each item in the navigation bar should have a clear and descriptive label that indicates the content of the page or section it links to. Avoid using jargon or abbreviations that might confuse users. For example, instead of using "Proj. Des.", use "Project Description."
- **Consistent Structure:** Maintain a consistent structure throughout the navigation bar. For example, if you use dropdowns for one category, ensure that all categories are similarly organized. This consistency helps users understand how to navigate your site intuitively.
- **Mobile Responsiveness:** Ensure that the navigation bar is responsive and functions well on mobile devices. This may involve using a hamburger menu for smaller screens, where users can click to expand the menu. Test the navigation on various devices to confirm usability.
- **Search Functionality:** If your wiki has a lot of content, consider incorporating a search bar in the navigation area. This allows users to quickly find specific information without having to browse through multiple pages.
- **Breadcrumb Navigation:** Implement breadcrumb navigation to provide users with context about their location within the site. This shows the path they took to arrive at a specific page, allowing them to backtrack easily.
- **Accessibility Features:** Ensure that the navigation bar is accessible to all users, including those with disabilities. Use appropriate color contrasts, keyboard navigation support, and ARIA (Accessible Rich Internet Applications) roles for assistive technologies.
- **Sticky Navigation:** Consider making the navigation bar sticky (fixed at the top of the page as the user scrolls). This keeps essential navigation options visible, improving user experience, especially on long pages.
- **Dropdown Indicators:** Use visual indicators (like arrows or plus signs) to show which menu items contain dropdown options. This helps users understand that more content is available under certain categories.
- **Highlighting Active Links:** Highlight active links (the current page) in a way that distinguishes them from other links. This could involve changing the text color or background color to indicate which page is currently being viewed.
- **Limited Number of Items:** Try to limit the number of top-level items in the navigation bar to avoid clutter. A clean, concise navigation menu is easier for users to navigate than one that is overloaded with links. If you have many categories, consider grouping them logically or using dropdown menus.
- **User Testing:** After designing the navigation bar, conduct user testing with team members or potential users to gather feedback on usability. This can help identify areas for improvement before finalizing the design.
- **Linking to External Resources:** If your project references external resources, ensure that links to these sites open in a new tab. This prevents users from losing their place on your wiki while exploring additional content.

By implementing these features in your navigation bar, you'll create a smoother and more intuitive browsing experience that can help judges and other users easily navigate your content and quickly access essential information.

4.5 Footer

Be sure to always include the obligatory parts in the footer such as the licensing information and link to you teams GitLab Repository.

- **Licensing Information:** It is essential to mention the licensing terms for your content in the footer.
- iGEM requires that all content be available under the Creative Commons Attribution 4.0 license or any later version, and this should be clearly stated.
- **GitLab Repository Link:** Include a visible link to your team's GitLab repository in the footer, as mandated by iGEM. This allows judges and future teams to easily access the source code used to generate the wiki.

Apart from the above parts, the footer design is quite flexible. Possible additions include:

- **Sponsor Links:** Many teams choose to include links to their sponsors in the footer, often displayed in a slider or as static images. This not only acknowledges support but also provides visibility for the sponsors.
- **Contact Information:** Some teams may include an Impressum (legal disclosure) and contact information in the footer. This typically consists of email addresses or physical addresses.
  - **Email Address Security:** If you choose to display email addresses, be mindful of security concerns. Simply listing an email address can expose it to spam bots. To mitigate this, consider using a contact form or disguise the email (e.g., using "teamname [at] domain [dot] com").
- **Navigation and Additional Links:** Consider including quick links to important pages or sections within the wiki in the footer, enhancing user navigation.
- **Social Media Links:** If applicable, teams may also choose to link to their social media profiles to engage visitors further and provide updates on their project.

By thoughtfully incorporating these elements into the footer, you can create a functional and informative section that enhances user experience while fulfilling the necessary requirements set forth by iGEM.

4.6 Header

Page headers are the first interaction point for users on your iGEM wiki. They should clearly display the page name and set the tone for the content. If you are aiming for best wiki, consider that engaging visuals, such as images or videos, can enhance the user experience while maintaining consistency with your team's branding and thoughtful headers create an inviting atmosphere that encourages exploration.

- **Clear Page Identification:** Every page header should clearly display the page name. This helps users quickly identify the content they are viewing and enhances navigability.
- **Visual Style:** Headers can vary in style, ranging from simple text to more elaborate designs. Some common styles include:
  - **Flashy Headers:** Utilizing bold graphics, vibrant colors, or animations to grab attention.
  - **Photographic Headers:** Incorporating relevant images that reflect the page content, creating an engaging visual experience.
  - **Video Headers:** Using short video clips or animations as headers can add dynamic content and enhance user engagement.
- **Call to Action:** If appropriate, headers can include calls to action (CTAs) to encourage users to engage further with the content, such as "Learn More" or "Get Involved."
- **Alignment with Content:** The style and imagery used in the header should align with the content of the page, providing context and setting the tone for what follows.

By thoughtfully designing page headers with these considerations in mind, you can create an inviting and informative entry point for users that enhances their overall experience on your iGEM wiki.

4.7 Additional Elements

Integrating extra elements into your iGEM wiki can significantly enhance navigation and user engagement. These features improve the overall experience by making content more accessible and encouraging exploration, creating a more inviting and informative environment for visitors.

- **Sidebar Navigation:** A sidebar can significantly improve navigation, especially for long pages. It should include only H1 and H2 headings, allowing users to quickly jump to different sections. Incorporating a highlight feature can indicate the current section the user is viewing.
- **"Back to Top" Button:** This button can be placed at the end of the page or within the sidebar, providing an easy way for users to return to the top without scrolling manually.

- **Cards:** Consider adding cards at specific positions, such as the end of each page, to engage users and encourage them to explore related content. These can highlight important links or additional resources.
- **Scrolling Progress Bar:** A visual scrolling progress bar can indicate how far the user has scrolled down the page, giving them a sense of navigation and encouraging continued reading.
- **Search Functionality:** Incorporating a search bar allows users to quickly find specific content or sections within the wiki, enhancing usability.
- **Suggested Reading:** Including a section for related links or suggested reading at the end of pages can guide users to additional relevant content, e.g. guiding the reader from design to engineering.

Final Thoughts

When structuring your iGEM wiki, prioritize user experience, especially from the perspective of judges and future teams. Keep navigation simple and intuitive, ensuring that all critical deliverables are well-documented and easy to locate. By maintaining clarity and organization across your pages, your team will be well-positioned for success in the competition.

Chapter 5

Getting started with the wiki

5.1 Getting an overview

You need to familiarise yourself with the requirements by iGEM and your team members. Some teams set high expectations for themselves in regards ot design or other aspects.

5.1.1 iGEMs resources

To successfully build your iGEM team Wiki, it is essential to thoroughly review iGEM's official Wiki Rules and Resources. Pay close attention to size limitations, time restrictions, and functionality constraints to ensure compliance with the competition requirements. Familiarise yourself with best practices and common pitfalls by examining previous Wikis, particularly those from past winning teams, as they can provide valuable insights into effective strategies and potential challenges.
Before you dive into coding, make sure to read iGEM's official information on the team Wikis and the GitLab Guide. These documents will help you understand the basic requirements, technical constraints, and available tools necessary for your Wiki's development. Key resources to consult include the iGEM GitLab Guide and the Team Wiki Deliverable page, which offer an overview of important documents that will guide you throughout the process.
By following these steps, you will be better prepared to create a well-structured and compliant Wiki that effectively showcases your team's project.

5.2 Creating the wiki team

Decide on one or multiple people taking the organisational lead for the wiki and determine which people will actively contribute to the wiki development. Everyone? A set group of people? Will everyone get a section to work on?
Depending on the answer you need to consider who will teach Further, consider who will manage coordination with other subteams.

5.2.1 Deciding on a template

The wiki lead needs to get an overview about the skills and resources available to your team. Based on the availability, you need to choose a template.

- **Markdown & Frozen Flask (Beginner)**: This simple template utilizes Markdown files to create a static website. It's user-friendly, making it an ideal choice for teams just starting out.
- **HTML & Frozen Flask (Intermediate)**: This template employs HTML files to generate a static website. While it is slightly more complex than the Markdown option, it remains relatively straightforward to use, making it suitable for teams with some coding experience.
- **React (Advanced)**: This option is for teams with advanced skills in React, a complex framework that allows for the creation of dynamic and interactive Wikis. Before choosing this template, ensure that your team is proficient in React, as we will not provide detailed guidance on its usage. It's crucial to review the requirements and thoroughly test your Wiki before the Wiki Freeze to ensure everything functions as intended.

5.3 Let everyone know the essentials

Everyone should be aware of the requirements and mistakes that could get you disqualified to avoid the responsibility being placed on one person. Important aspects are:

- GitLab Repository Link
- Licensing
- Standard URL Pages
- Content Hosting
- iframes Usage
- Source Code Submission

Do not rely on the knowledge of former teams only as the guidelines can change.

5.4 Review tasks and timeline

Get started with the following questions:

Who will oversee

What pages do we need?

How do we distribute

5.5 Distributing responsibility

To effectively build your iGEM team Wiki, it's essential to clearly define team roles and responsibilities. Consider assigning specific positions such as content creators, developers, and designers. Utilizing task management tools can help streamline the process and ensure everyone stays organized.
Each team member should be aware of their specific tasks related to the Wiki, which may include:

- **Page Layouts and Design:** Creating visually appealing and user-friendly layouts.
- **Coding:** Implementing the necessary code for functionality.
- **Automation:** Streamlining repetitive tasks where possible.
- **Data Curation:** Organizing and managing data sets relevant to your project.
- **Media Management:** Uploading videos, taking and editing photographs (including considerations for lighting, cropping, and naming conventions).
- **Text Creation and Editing:** Writing, editing, and correcting text to ensure clarity and accuracy.
- **Task Management:** Overseeing progress and reminding team members of their responsibilities.
- **Mobile Optimization:** Ensuring the site is accessible on mobile devices.
- **Illustrations and Animations:** Creating visual elements to enhance the content.
- **Accessibility:** Implementing features that make the Wiki usable for all visitors.
- **Documentation:** Maintaining accurate records of processes and changes.
- **Development of Additional Tools:** Creating helpful scripts or tools, such as Python scripts, to automate tasks.

By clearly defining roles and responsibilities and utilizing task management tools, your team can effectively collaborate to create a comprehensive and engaging Wiki for the iGEM competition. It is recommended to distribute these tasks on multiple people.
Most tasks concerning the Wiki overlap with other subteams and should therefore be handled in cooperation with these.

Chapter 6

File structure

To facilitate collaborative work on the wiki and minimize merge conflicts in GitLab, it's essential to establish a well-organized file structure. This will help multiple team members work simultaneously without issues, especially for lengthy sites.

Organized Components: For pages with multiple tabs or sections, create individual files for each tab and treat them as components. This modular approach enhances clarity and maintainability.

Suggested Folder Structure

- `project-folder/`
  - `code/`: Additional tools, e.g. Python scripts
  - `public/`: Files you need to be able to access with via url, e.g. java scripts
  - `src/`
    - `app/`: Store the App and other main components.
    - `components/`: Reusable components like buttons, cards, and sections.
    - `data/`: Data sets used for automated components or other elements.
    - `ic/`: ?? and ?? files. Alternatively, you can use a global definition file.
    - `pages/`: The files used as page components.
    - `styles/`: CSS or SCSS files to manage styling consistently.
    - `utils/`: Type Script and JavaScript functions for global use such as ?? or ??.
    - `main.tsx`
    - `navigation.ts`
    - `pages.ts`
    - `index files`
  - `README`
  - `??`
  - `LICENSE`
  - `index.html`
  - `gitignore`
  - `pipeline file`

This structure can be extended to, for example, include files for headers, sidebars and references per page.

6.1 Naming Conventions

Implement clear and consistent naming conventions for files and folders to enhance navigation and understanding within the project.

- **PascalCase** for props, states and components: *UserProfile, Navbar, ItemList.*
- **camelCase** for variables: *wordCount, primaryColor*
- **Verb-Noun with context** for functions: *UseNavbar, CreateSidebar*
- **Lowercase with Hyphens** for classes and ids: *experiment-header, first-button*
- **Context** for all naming: *Timeline, InfoBoxes, createSidebar*

Chapter 7

Media - Pending

Upload pending.

Chapter 8

Wiki guide for team - Pending

Upload pending.

Wiki development

Chapter 9

Coding prequisites

9.1 Hardware

The most obvious requirement is a laptop or computer. In my experience, laptop-tablet 2-in-1 devices can be difficult to work with due to limitations in handling coding tasks. Additionally, not all laptops can smoothly run the required coding environments. We have encountered common issues with Windows laptops, particularly around scripts not being allowed to run or installations being blocked. On personal devices, these problems can usually be fixed (with an internet search), but on university-provided laptops, users may not have the administrative rights needed to make necessary changes.

I recommend to address these issues on debut, as deferred tasks tend to not get done in iGEM projects and trying to fix them last minute may lead to delays that are difficult to recover from.

9.2 GitLab Access and Cloning the Repository

To contribute to the Wiki, every team member needs to clone the repository from the iGEM GitLab. This requires each member to have their iGEM account credentials (username and password) ready. I recommend using SSH cloning for a more secure and stable connection.

HTTPS Cloning: This method requires users to enter their iGEM credentials (username and password) every time they push or pull code. It is straightforward but can become annoying due to repeated logins.

- **Advantages:** Easy to set up and doesn't require additional software.
- **Disadvantages:** Requires entering credentials frequently, less secure than SSH.

SSH Cloning: This method uses an SSH key to authenticate, avoiding the need to repeatedly enter a username and password. It is generally considered more secure and efficient.

- **Advantages:** No need for constant credential input, more secure, ideal for frequent use.
- **Disadvantages:** Requires setting up an SSH key and ensuring the necessary software is installed.

Setting Up SSH Cloning:

For SSH cloning, each team member needs an **SSH key generator** installed. You can check if your system has this by typing `ssh` in the terminal. If the system does not recognize the command, you'll need to install an SSH key generator that fits your operating system.

It is best to explain to the team what an SSH key is: it is a pair of cryptographic keys used for secure access. Always remember to **only use and share the public key**, never the private key. The public key typically has the file extension `.pub`.

> **Note:** On some devices, `.pub` files may attempt to open with Microsoft Publisher or produce an error about missing software. You can simply open the file using the "Open with..." option and choose a text editor.

9.2.1 Step-by-Step explanation

1. **Check for SSH Key Generator:** Open your terminal and type `ssh` to check if the key generator is installed. If not, install it (e.g., OpenSSH for Windows or macOS/Linux).
2. **Generate the SSH Key:** In the terminal, enter the command to generate the SSH key (e.g. `ssh-keygen`).
   You will be asked:

   ```
   Enter file in which to save the key (/home/user/.ssh/id_ed25519):
   ```

   Press **Enter** to accept the default file name or enter a different name to create a new key if the file already exists.
3. **Set a Passphrase (Optional):**
   At the prompt:

   ```
   Enter passphrase (empty for no passphrase):
   ```

   Choose to set a passphrase or press **Enter** to skip. You will be prompted to confirm the passphrase.
4. **View Key Information:** After generating the key, the terminal will display the file name, its fingerprint, and a randomart image of the key.
5. **Log into iGEM GitLab:** Go to iGEM GitLab and log in with your credentials.
6. **Access SSH Key Settings:**
   - Click on your profile picture (top right) and select **Preferences**.
   - This will take you to the **User Settings** page.

7. **Add New SSH Key:**
    - On the left, click **SSH Keys**.
    - Then, click **Add new key**.
8. **Copy Your Public Key:**
    - Open the public key file you just generated (it typically ends with `.pub`).
    - *Note*: You may need to reveal hidden files to access the `.ssh` folder.
    - Copy the entire content of the public key file. It will start with `ssh-ed25519` or `ssh-rsa`.
9. **Paste the Key in GitLab:**
    - Paste the public key content into the **Key** box on GitLab.
    - Add a **Title** (e.g., "My Laptop Key").
    - Click **Add key** to save it.

After completing these steps, your SSH key will be added, and you can now clone repositories via SSH!
For that, install the necessary software as described in the following section.

9.3 Coding Environment (IDE) and Git

I recommend every team member installing a coding environment (IDE) to work efficiently. Popular choices include `Visual Studio Code` or `IntelliJ IDEA`.

While it's possible to edit files directly on GitLab through:

- **The Text Editor:** No helper tools to catch errors.
- **The Web IDE:** Limited error-checking capabilities.

A standard IDE provides (better) error detection, auto-completion, and other productivity tools that make coding more efficient and less error-prone.

But an IDE alone does not suffice. To start working with Git, you'll need to install it on your system and configure it with your username and email, which will identify your contributions.

1. Install Git : Either download the installer from https://git-scm.com/downloads and follow the setup instructions depending on your operating system or google a way to install it via the command line.

2. Set Up Username and Email : After installing Git, configure your username and email to be used in your commits. This ensures that every commit you make is properly attributed to you.

1. Open a terminal or command prompt.
2. Set your username by running:

```
git config --global user.name "Your Name"
```

3. Set your email address by running:

```
git config --global user.email "your.email@example.com"
```

You can check if your Git username and email are set correctly by running:

```
git config --global --list
```

9.4 Cloning your repository

After installing Git and an IDE, you are ready to clone your repository.

1. Go to Your Project Page on GitLab: Navigate to the iGEM GitLab project page you want to clone.

2. Open the Code Menu: On the project page, click on the blue **Code** button near the top of the screen.

3. Select "Open in Your IDE": From the dropdown, select **Open in your IDE** for simplicity. You will be presented with options to select between different IDEs. Choose either:

- **Visual Studio Code (SSH)**
- **IntelliJ IDEA (SSH)**

4. Choose a Folder: A file explorer will open, prompting you to select a folder on your local machine where the repository will be saved. I advise against saving the folder in a cloud. Git already backs your work up, if you push (upload it).

Trust the Authors (if prompted): If a pop-up appears asking whether you trust the project's authors, confirm by selecting **Trust** to ensure you can edit the files without restrictions.

9.5 Other Software

Depending on the type of project in your repository, you will need to install specific package managers and dependencies. For React projects, some of those will already be listed in your **??**s, additional ones you need to install as the necessity arises.
This will be covered in React.

Additional Considerations

Regardless of the project type, be aware that there may be additional packages, scripts or software required depending on what is already installed on your device. The error messages you encounter will usually indicate what is missing.

Further, you might need to think about:

- **Using the Correct Shell:** Some commands might differ based on the shell you're using (e.g., bash, zsh, cmd, PowerShell). Ensure you are using the correct commands for the terminal of your operating system.
- **using terminals as an administrator** Some changes or installations can only be made if you do them as an administrator. It can differ from operating system to operating system how you install things as an administrator.
- **Common Dependency Errors:** If you encounter errors related to dependencies, search for the specific error message online, as there may be community solutions or discussions about the issue.
- **File Permissions:** On some operating systems, particularly Linux and macOS, you may need to adjust file permissions for certain operations. That does not automatically mean you should not do the operation.

Chapter 10

General Coding

To be able to contribute to the iGEM Wiki, it is beneficial for all team members to have a basic understanding of **HTML** and for at least the wiki team members to have a good understanding of **HTML** and a basic understanding of **CSS** . Familiarity with these languages will enable everyone to assist with content creation, formatting, and layout adjustments.

Consider organizing internal wiki workshops early in the season to teach the basics of HTML and CSS. This lowers the entry barrier for new members and enables more people to contribute to content creation and formatting.

10.1 HTML

HTML is used to define the structure and content of a webpage. It determines elements such as headings, paragraphs, tables, images, links, and sections. A clear and semantic HTML structure improves readability, maintainability, and accessibility.

There are many free resources available online, for example:

- `https://www.w3schools.com/html/default.asp`
- `https://www.geeksforgeeks.org/html/html-cheat-sheet/`

If you are in a hurry, here are some key concepts to get started:

- **Opening and Closing Tags**: Each HTML element typically consists of an opening tag and a closing tag. For example, in `<p>Your text here</p>`, `<p>` is the opening tag, and `</p>` is the closing tag.
- **Nesting of Tags**: HTML tags can be nested within one another, meaning that one tag can contain another. For instance, `<div><p>Your text here</p></div>`.

- **Semantic HTML**: Using appropriate tags for their intended purpose (like `<header>`, `<footer>`, `<article>`, etc.) improves accessibility and SEO.
- **Attributes**: HTML elements can have attributes that provide additional information about them. For example, the `<a>` tag uses the `href` attribute to specify the link's destination: `<a href="https://example.com">Link</a>`.

10.2 CSS and SCSS

CSS is used to style and arrange HTML elements. It controls colors, fonts, spacing, layouts, animations, and responsive behavior for different screen sizes. Even a basic understanding of CSS allows team members to make quick design adjustments and improve the overall appearance of the wiki.

Difference Between CSS and SCSS

SCSS (Sassy CSS) is a syntax of SASS (Syntactically Awesome Style Sheets), which extends CSS with features like variables, nested rules, and mixins. While CSS allows for styling web pages, SCSS offers more advanced capabilities for organizing and writing styles, making it easier to manage complex stylesheets.

How HTML and CSS Work Together

HTML provides the structure and content of a webpage, while CSS handles the visual presentation. HTML elements are styled using CSS selectors, allowing developers to specify how different components should look. For example, a `<h1>` element can be styled to change its font size, color, and alignment using CSS.
Resources:

- `https://www.w3schools.com/css/default.asp`
- `https://www.geeksforgeeks.org/css/css-complete-guide/`
- `https://www.w3schools.in/scss/introduction`

Basic concepts for a quick start:

- **Selectors**: There are different types of CSS selectors (element, class, ID). For example, `.classname` targets all elements with that class, while `#idname` targets a unique element.

- **Box model**: The box model describes how elements are rendered and how spacing around them is calculated, influencing overall layout. E.g. the aspects content, padding, border or margin
- **Flexbox and grid**: These modern CSS layout techniques help create responsive designs.
- **Responsive design through media queries**: Media queries enable styles to adapt based on screen size, orientation, or resolution, providing an optimal user experience across devices.

Personal Variables

Setting personal colors and variables in CSS is a good way to ensure design consistency, enhance maintainability, and make your styles easier to modify across an entire project. You can centralize design elements like colors, sizes, and gradients, making it simpler to update and reuse them throughout your code.

In CSS, custom properties are defined using the `- -` prefix. For example, you might want to set up a collection of color variables that represent your project's brand or theme. These could include primary and secondary colors, as well as different shades for various purposes.

```
:root {
  --text-primary: #850F78;
  --mediumpurple: #bc15aa;
  --lightpurple: #B85BD1;
  --very-light-purple: #ce9fc9;
}
```

In this example, different shades of purple are defined as variables. The `:root` selector ensures that these variables are globally available throughout your CSS. Now, instead of repeating color codes in different parts of your stylesheet, you can reference these variables:

```
h1 {
  color: var(--text-primary);
}

.button {
  background-color: var(--mediumpurple);
}

.highlight {
  border-color: var(--lightpurple);
}
```

You can also use CSS variables to define more complex styles, such as gradients or sizes.

```
:root {
  --node-size: 60px;
  --ourgradient: linear-gradient(90deg, rgba(147,94,184,1) 0%, rgba
     (160,167,243,1) 100%);
}
```

Here, `--node-size` defines a consistent size that can be applied to elements like buttons, icons, or divs, while `--ourgradient` provides a linear gradient that can be used as a background for various components.

```
.icon {
  width: var(--node-size);
  height: var(--node-size);
}

.header {
  background: var(--ourgradient);
}
```

10.3 JavaScript and TypeScript

While HTML and CSS define the structure and appearance of a webpage, **JavaScript** adds functionality and interactivity. It allows websites to react to user input, update content dynamically, create animations, validate forms, or load data without reloading the page.

JavaScript can be used to create interactive diagrams, navigation menus, expandable content sections, image galleries, animated statistics, or custom user experiences. Even small scripts can significantly improve usability and presentation quality.

TypeScript is an extended version of JavaScript that adds static typing and improved tooling. It helps prevent programming errors, improves code readability, and becomes especially useful when several people collaborate on the same project. For larger wiki projects, TypeScript is often preferable to plain JavaScript. My template uses TypeScript

Relevant differences are:

- **Typing**: JavaScript is dynamically typed, meaning variables can hold any data type, and type checking occurs at runtime. TypeScript is statically typed, allowing for type annotations and catching errors during development.

- **Development Experience**: TypeScript offers better tooling support, including autocompletion, type checking, and easier refactoring, leading to a more robust development experience.
- **Compilation**: TypeScript must be transpiled (compiled) to JavaScript before it can run in the browser, while JavaScript can be executed directly.
- **Language Features**: TypeScript supports modern JavaScript features and introduces additional functionalities not present in standard JavaScript, like interfaces and decorators.

If the team has little prior programming experience, it is perfectly reasonable to begin with JavaScript and later transition to TypeScript when the project grows in complexity.

Useful learning resources include:

- `https://www.w3schools.com/typescript/index.php`
- `https://www.w3schools.com/js/default.asp`
- `https://www.geeksforgeeks.org/typescript/typescript-tutorial/`
- `https://www.geeksforgeeks.org/javascript/javascript-tutorial/`

Chapter 11

React

11.1 Setting up your React project

Updating Package Managers: If you already have `npm` or `yarn` installed, consider updating them to the latest version to avoid compatibility issues. You do not need both package managers, one should be sufficient. If you are unsure, read about the differences between the managers to make a decision. Otherwise **Install Package Mangers:**

- **node**: Make sure you have Node.js installed. You can check by running `node --version`. If it's not installed, download it from the Node.js website.
- **npm:** It should come bundled with Node.js.
- **yarn**: If you don't already have yarn installed, you can usually do so by running the following command:

```
npm install --global yarn
```

Clone your project if you have not already done so (see Cloning your repository). If you want to use my template, simply download the project from `https://github.com/liliana-sanfilippo/template-wiki` as a zip file and move or copy everything (or the desired parts if you know what you are doing) into the root folder of your cloned project. If you do not know what the root directory is, see here: `https://www.geeksforgeeks.org/git/how-to-navigate-to-the-git-root-directory/`.

Install Project Dependencies: Navigate to your project directory in the terminal and run:

```
yarn install
```

or

```
npm install
```

Ensure Compatible Node.js Version: Make sure your Node.js version is either `^8.18.0` or `>=20.0.0` to avoid compatibility issues, with `18.20.0` being the recommended version in 2024. This is a recommendation for using my template wiki and software and does not apply to all projects.

11.2 Learning React

There are already many resources for React so there is no use for me to copy what they say here. I recommend w3schools again: `https://www.w3schools.com/react/react_intro.asp` to learn React basics and Geeks for Geeks for people wanting to get deeper into React: `https://www.geeksforgeeks.org/reactjs/react/`.

If you just want to write texts and put them on the wiki, please learn about:

- JSX: `https://www.w3schools.com/react/react_jsx.asp` or `https://www.geeksforgeeks.org/reactjs/reactjs-jsx-introduction/`
- JSX styling/attributes; `https://www.w3schools.com/react/react_jsx_attributes.asp`

If you want to create simple components, please additionally learn at least about:

- JSX If statements: `https://www.w3schools.com/react/react_jsx_if_statements.asp`
- React components: `https://www.w3schools.com/react/react_components.asp` or `https://www.geeksforgeeks.org/reactjs/reactjs-components/`
- React Props: `https://www.w3schools.com/react/react_props.asp`, `https://www.w3schools.com/react/react_props_destructuring.asp`, `https://www.w3schools.com/react/react_props_children.asp`
-

And ideally as well about:

- React Conditional Rendering: `https://www.w3schools.com/react/react_conditional_rendering.asp` or `https://www.geeksforgeeks.org/reactjs/reactjs-conditional-rendering/`

For components that work with data sets (e.g. automatic profile rendering for the team page) specifically also:

- Mapping `https://www.w3schools.com/react/react_lists.asp`

If you want to import styling either generally into the App or into specific components:

- `https://www.w3schools.com/react/react_css_modules.asp`

If you want to alter the template or create more complex components, please learn about:

- React Hooks: `https://www.w3schools.com/react/react_hooks.asp`
- React Router (if applicable): `https://www.w3schools.com/react/react_router.asp`

There are many useful libraries and packages for styling or components.

- Styling and components: `https://www.geeksforgeeks.org/reactjs/react-bootstrap/`
- Styling: `https://www.geeksforgeeks.org/css/tailwind-css/`

Chapter 12

Additional tools - Under construction

12.1 Bootstrap

Bootstrap is a front-end framework that simplifies the process of designing responsive and mobile-first websites. It provides a collection of pre-designed HTML, CSS, and JavaScript components, such as buttons, forms, navigation bars, and modals.

You could use Bootstrap for:

- **Responsive Grid System**: Create a multi-column layout for a product listing page that adjusts to 2 columns on tablets and 4 columns on desktops using Bootstrap's grid classes like `.col-md-6` and `.col-lg-3`.
- **Navigation Bar**: Implement a sticky navigation bar at the top of your web page that collapses into a hamburger menu on mobile devices, using Bootstrap's `.navbar` and `.navbar-toggler` classes.
- **Image Carousel**: Add an image slider on your homepage to showcase featured products or testimonials, utilizing Bootstrap's carousel component with classes like `.carousel`, `.carousel-item`, and `.active`.
- **Cards for Content Display**: Design a series of content cards to present team member profiles, each containing a photo, bio, and social media links, utilizing Bootstrap's card component.
- **Buttons with Icons**: Create stylish buttons for social media sharing that include icons, by combining Bootstrap's button classes with Font Awesome icons.
- **Tooltips for Additional Information**: Add tooltips to icons or buttons that provide extra context when hovered over, using Bootstrap's tooltip component for improved user experience.
- **Responsive Tables**: Create a responsive data table that allows users to scroll horizontally on smaller screens, using Bootstrap's table classes like `.table` and `.table-responsive`.
- **Footer with Social Links**: Design a footer that includes links to your social media profiles with Bootstrap's utility classes to align items properly.

Chapter 13

Troubleshooting - Pending

Upload pending.

Chapter 14

Accessibility - Under Construction

Upload pending.

In the meantime: https://static.igem.wiki/teams/5172/pdf/accessibility-guide.pdf and https://static.igem.org/mediawiki/2021/9/93/T--TU_Darmstadt--guideline-web-accessibility.pdf

Chapter 15

Scientific writing - Under Construction

Upload pending.

In the meantime: `https://static.igem.org/mediawiki/2019/8/89/T--Wageningen_UR--Wiki_writing_style_guide.pdf`

Chapter 16

Other frameworks and systems - Under Construction

Upload pending.

In the meantime:

- WordPress: `https://static.igem.org/mediawiki/2021/e/e2/T--TU_Darmstadt--gogem-how-to-wiki-the pdf`
- HTML: `https://www.w3schools.com/html/`
- Rust + Dioxus: `https://2024.igem.wiki/marburg/how-to-wiki`
- Jekyll: `https://2024.igem.wiki/sdu-china/wiki-code`
- Markdown:
  - `https://2024.igem.wiki/shanghaitech-china/docs/writing/` or rather `https://2024.igem.wiki/shanghaitech-china/docs/`
  - `https://igem-wiki-starter.readthedocs.io/en/latest/`
  - `https://github.com/Lambert-iGEM-2022/Wiki-Instructions`
  - `https://github.com/DKSA1/igem-wiki-starter`
  - (chinese) `https://2021.igem.org/Team:Fudan/WikiInstruction`

Chapter 17

The Wiki Freeze - Under construction

17.1 Internal freeze approach

For both teams I was part of, it was helpful to have an internal “freeze day” in the form of a working session where everything that could and should be done is checked. Afterward, only new results should be uploaded, no design changes should be made, etc. Alumni and professors can be invited to give input, proofread texts or help otherwise.

Special prizes

17.2 Special prize page

You can create a dedicated special prize page where you outline how you fulfilled the requirements.

Chapter 18

Best Wiki - Pending

Upload pending.

After the Jamboree

Chapter 19

The Wiki Thaw - Pending

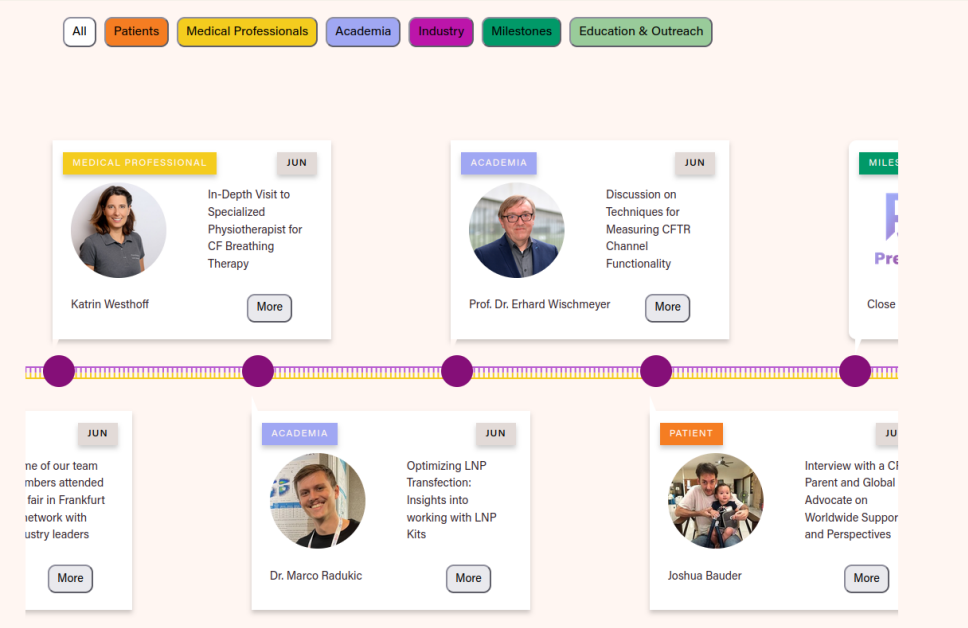
Upload pending.

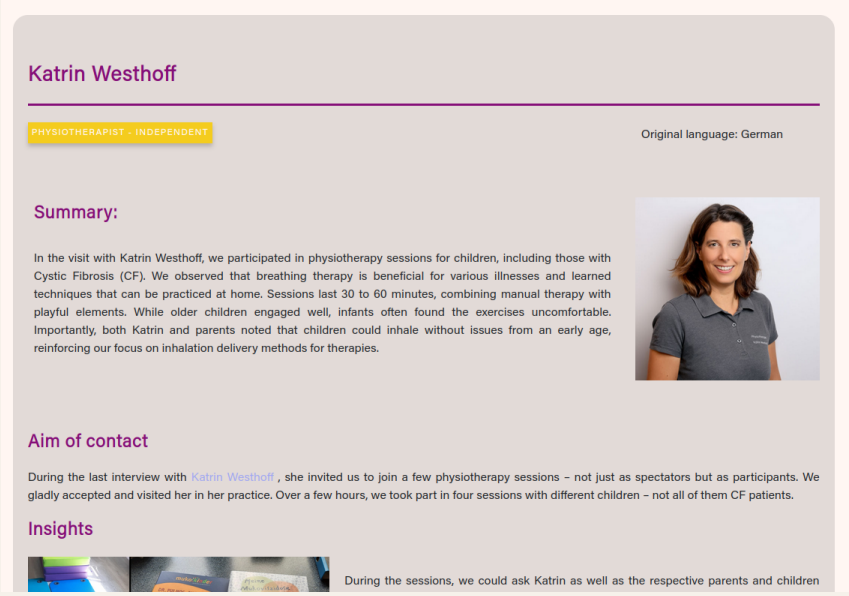
Documentation framework examples

19.1 Human Practices Documentation Framework

19.1.1 Timeline - Bielefeld-CeBiTec 2024 (Best IHP Winner)

The implementation





Katrin Westhoff

PHYSIOTHERAPIST - INDEPENDENT

Original language: German

Summary:

In the visit with Katrin Westhoff, we participated in physiotherapy sessions for children, including those with Cystic Fibrosis (CF). We observed that breathing therapy is beneficial for various illnesses and learned techniques that can be practiced at home. Sessions last 30 to 60 minutes, combining manual therapy with playful elements. While older children engaged well, infants often found the exercises uncomfortable. Importantly, both Katrin and parents noted that children could inhale without issues from an early age, reinforcing our focus on inhalation delivery methods for therapies.

Aim of contact

During the last interview with Katrin Westhoff , she invited us to join a few physiotherapy sessions – not just as spectators but as participants. We gladly accepted and visited her in her practice. Over a few hours, we took part in four sessions with different children – not all of them CF patients.

Insights

During the sessions, we could ask Katrin as well as the respective parents and children

The structure behind it

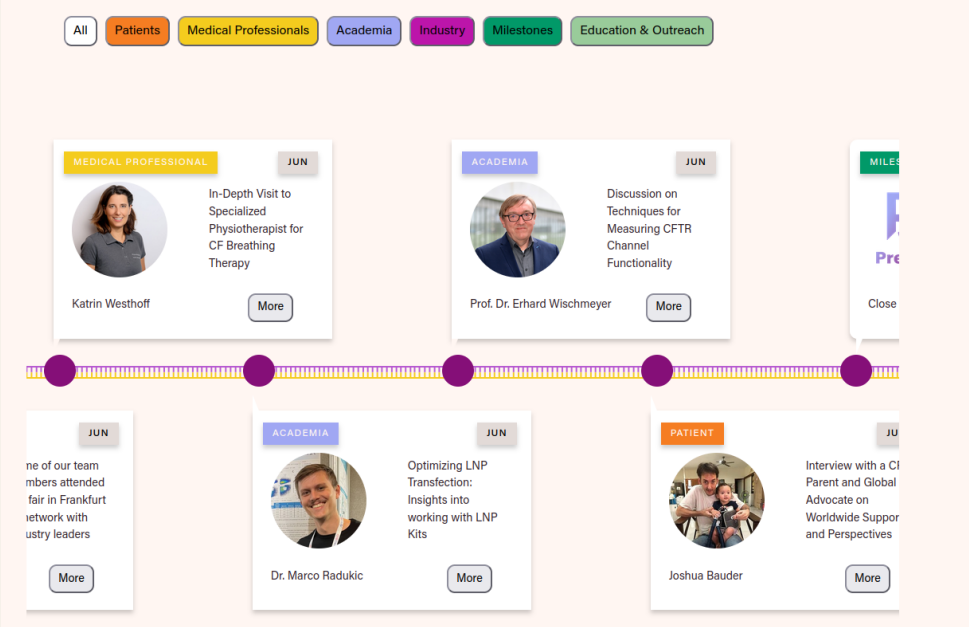
```
export interface TimelineDatenpunkt {
  title?: string; /* Prof. , Dr., ... */
  vorname: string;
  nachnname: string;
  pictureurl: string;
  tag: StakeholderTag;
  heading: string;
  interviewtabid: string;
  summary: string | Array<string> | Array<React.ReactNode>; /* Array<React.
     ReactNode> for anything that needs HTML, otherwise strings */
  months: string;

  type?: TypeTag; /* only if it is a meta tag */
  affiliation?: string;
  job?: string;
  language?: Language;
  quote?: string;
  quoteVorname?: string; /* If the quote is not from the person the text is
     about */
  quoteNachname?: string;
  aimofcontact?: string | Array<string> | Array<React.ReactNode>; /* Array<
     React.ReactNode> for anything that needs HTML, otherwise strings */
  insights?: string | Array<string> | Array<React.ReactNode>; /* Array<
     React.ReactNode> for anything that needs HTML, otherwise strings */
  clarification?: string | Array<string> | Array<React.ReactNode>; /* Array
     <React.ReactNode> for anything that needs HTML, otherwise strings */
```

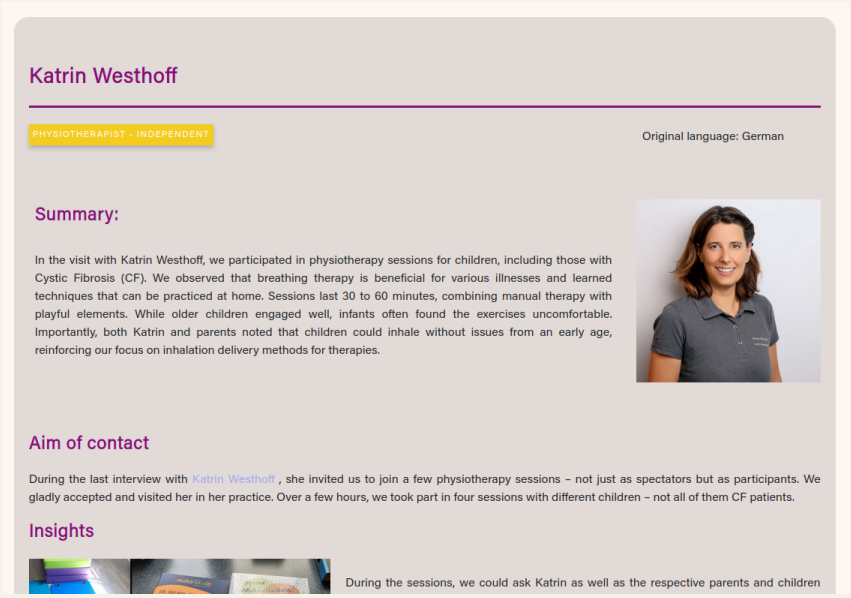
```
  implementation?: string | Array<string> | Array<React.ReactNode>; /*
     Array<React.ReactNode> for anything that needs HTML, otherwise strings
      */
  pictureurl_interview?: string;  /* Picture that goes into the paragraph "
     Insights"  */
  pictureurl_aim?: string;  /* Picture that goes into the paragraph "Aim of
      contact"  */
  pictureurl_implementation?: string;  /* Picture that goes into the
     paragraph "Implementation"  */
  more_pictures?: Array<string> ;
  references?: React.ReactNode;  /* Has to be HTML Code - we were not using
      the citation component yet */
  interview?: React.ReactNode;
  text?: string | Array<string> | Array<React.ReactNode> |React.ReactNode ;
      /* any extra text */
  experton?: string;
}
```

The timeline component rendered visual error messages if aspects were missing. This approach led to the team only having to fill out the data sheet and getting visual feedback for missing info. No HTML coding required, but possible.

Adaptation by DTU Denmark 2025 (Top 10, Best Wiki Nominee)

See at `https://2025.igem.wiki/dtu-denmark/human-practices`


All
Patients
Medical Professionals
Academia
Industry
Milestones
Education & Outreach
MEDICAL PROFESSIONAL
JUN
In-Depth Visit to Specialized Physiotherapist for CF Breathing Therapy
Katrin Westhoff
More
ACADEMIA
JUN
Discussion on Techniques for Measuring CFTR Channel Functionality
Prof. Dr. Erhard Wischmeyer
More
Close
JUN
ne of our team
mbers attended
fair in Frankfurt
etwork with
ustry leaders
More
ACADEMIA
JUN
Optimizing LNP Transfection: Insights into working with LNP Kits
Dr. Marco Radukic
More
PATIENT
Interview with a CF Parent and Global Advocate on Worldwide Suppor and Perspectives
Joshua Bauder
More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Insights
During the sessions, we could ask Katrin as well as the respective parents and children

Associated software

Bibtex Parser
COMPLETED · package – script– component
Webpack parser script usable as TypeScript and React component
`https://github.com/liliana-sanfilippo/bibtex-ts-parser`

Author names Parser
IN PROGRESS · package – script– component
Name parsing grammar for ANTLRts, indended to be used to parse names of authors in bibtex sources and webpack parser script usable as TypeScript and React component.
`https://github.com/liliana-sanfilippo/author-name-parser`

Reference generator component
IN PROGRESS · package – component
TypeSxcript and React component using bibtex and nam parser to automatically parse references for wiki pages or scientifc blogging.
`https://github.com/liliana-sanfilippo/react-bibtex-source-generator`

React Library
IN PLANNING · package – component
Library of TypeScript and React components aimed at iGEM wikis.
`https://github.com/liliana-sanfilippo/igem-react-lib`

Attributions

LaTeXcode

Layout	**Rafael Kümmel** CC BY 4.0 · Apostila Template CP-USP-2021 Overall document layout was adapted from Kümmels template though the code changed and added to. `www.overleaf.com/latex/templates/apostila-template-cp-usp-2021/tfcwfrsgzhbz`
Attribution table	**Sarah Lang** CC BY 4.0 · Modern Simple CV Table layout was adapted. `https://www.overleaf.com/latex/templates/modern-simple-cv/kwrxbwthgrwr`

Illustrations

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Simple Mitochondria Rod
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Mitochondrium 6
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